Java Data Structures Library Project
Project Requirements

Minimum requirements

Any items marked incomplete in this section will result in the project receiving a failing grade.

	Complete?
Binary math	
User can input a decimal number and convert it to binary. The binary number should be displayed in the CLI.	
User can input a binary number and convert it to decimal. The decimal number should be displayed in the CLI.	
User can input two binary numbers and add them together. The sum should be displayed in the CLI.	
User can input two binary numbers and subtract them. The difference should be displayed in the CLI.	
User can input two binary numbers and multiply them. The product should be displayed in the CLI.	
User can input two binary numbers and divide them. The quotient should be displayed in the CLI.	
Stack	
User can push an item onto the stack. The stack should be displayed in the CLI.	
User can pop an item off the stack. The stack should be displayed in the CLI.	
User can peek at the top item of the stack. The item should be displayed in the CLI.	
User can check if the stack is empty. A message should be displayed in the CLI.	
User can check if the stack is full. A message should be displayed in the CLI.	
User can clear the stack. The stack should be displayed in the CLI.	
Queue	
User can add an item to the queue. The queue should be displayed in the CLI.	
User can remove an item from the queue. The queue should be displayed in the CLI.	
User can peek at the front item of the queue. The item should be displayed in the CLI.	
User can check if the queue is empty. A message should be displayed in the CLI.	
User can check if the queue is full. A message should be displayed in the CLI.	
User can clear the queue. The queue should be displayed in the CLI.	
Linked list	
User can add an item to the linked list. The linked list should be displayed in the CLI.	
User can remove an item from the linked list. The linked list should be displayed in the CLI.	
User can search for an item in the linked list. A message should be displayed in the CLI.	
User can check if the linked list is empty. A message should be displayed in the CLI.	
User can clear the linked list. The linked list should be displayed in the CLI.	
User can display the linked list in reverse order. The linked list should be displayed in the CLI.	
User can display the linked list in ascending order. The linked list should be displayed in the CLI.	
User can display the linked list in descending order. The linked list should be displayed in the CLI.	
Overall requirements	
The app is deployed online so that the rest of the world can use it.	
You are shown as the only contributor to the project on GitHub.	

Convention requirements

More than two items marked incomplete in this section will result in the project receiving a failing grade.

	Complete?
The files in the app are organized following the conventions demonstrated in lectures.	
The code in the app adheres to coding conventions covered in lessons, like using plural names for arrays.	
The app does not contain dead code, commented-out sections, or console logs.	
The app can be used without encountering errors.	
The app is coded using proper indentation.	
The app's menu is easily navigable by a first-time user.	
The GitHub repository used for the project must be named appropriately. For example, names that describe your project are appropriate, whereas generic names such as ga-project or unit-project are not.	

README requirements

More than two items marked incomplete in this section will result in the project receiving a failing grade.

	Complete?
The app's name.	
A brief description of your app and its functionality.	
A **Getting started** section. This section should include instructions on how to use your app.	
A **Technologies used** section. This section should list the principal technologies used by your application, including any major frameworks or libraries.	
If necessary, an **Attributions** section. This section should include links to any external resources (such as libraries or assets) you used to develop your application that require attribution. You can exclude this section if it does not apply to your application.	

Summary

The project meets the requirements above and passes.	